MAST32001 Computational Statistics - Lecture 14

Knockoffs, Variable Selection and False Discovery Rate

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Motivation

Variable selection creates a multiplicity problem

Suppose

$$Y = f(X_1, \ldots, X_p) + \varepsilon.$$

We want to discover which variables are associated with the response.

If we inspect many variables, some will appear important purely by chance.

The computational challenge is therefore not only

find strong variables,

but

find strong variables while controlling false discoveries.

Why ordinary variable importance is not enough

A fitted model can rank variables by:

- t-statistics;
- lasso entry times or coefficients;
- likelihood changes;
- random-forest importance;
- another predictive score.

But a large score alone does not tell us how many selected variables are false discoveries.

Knockoffs create synthetic competitors that act as **negative controls**.

Multiple testing and FDR

Selection notation

Suppose there are p candidate variables, indexed by

$$\{1, \ldots, p\}.$$

Let $\mathcal{H}_0$ denote the unknown set of **null variables** and let

$$\mathcal{H}_1 = \{1, \ldots, p\} \setminus \mathcal{H}_0$$

denote the non-null variables.

A variable-selection procedure returns a random selected set

$$\widehat{S} \subseteq \{1, \ldots, p\}.$$

The word *random* matters: $\widehat{S}$ changes when the observed data change.

Discoveries and false discoveries

For one realized data set, define

$$R = |\widehat{S}|$$

as the total number of discoveries, and

$$V = |\widehat{S} \cap \mathcal{H}_0|$$

as the number of false discoveries.

Thus $V \leq R$. The difficulty is that $\mathcal{H}_0$, and therefore V, is unknown in real data.

False discovery proportion

For one realized analysis, the **false discovery proportion** is

$$\mathrm{FDP} = \frac{V}{R \vee 1}, \qquad R \vee 1 = \max(R, 1).$$

The $\vee 1$ convention avoids division by zero when no variables are selected.

If $R > 0$, FDP is simply the fraction of selected variables that are actually null. If $R = 0$, FDP is defined to be zero.

False discovery rate

The **false discovery rate** is

$$\mathrm{FDR} = E(\mathrm{FDP}) = E\left[\frac{|\widehat{S} \cap \mathcal{H}_0|}{|\widehat{S}| \vee 1}\right].$$

The expectation is over repeated realizations of the data and any randomness in the selection procedure.

A procedure controls FDR at level $q \in (0, 1)$ if

$$\mathrm{FDR} \leq q.$$

This is an average repeated-sampling guarantee; it does **not** say that every realized selected set has FDP at most q.

FWER versus FDR

The **family-wise error rate** is

$$\mathrm{FWER} = P(V \geq 1),$$

the probability of making at least one false discovery.

The **false discovery rate** is

$$\mathrm{FDR} = E\left(\frac{V}{R \vee 1}\right),$$

the expected proportion of false discoveries among the selected variables.

FWER targets the event of *any* false discovery. FDR targets the *proportion* of false discoveries and can therefore be less restrictive when many discoveries are sought.

What must a knockoff method accomplish?

We would like to construct a selected set $\widehat{S}$ while controlling

$$E\left[\frac{|\widehat{S} \cap \mathcal{H}_0|}{|\widehat{S}| \vee 1}\right].$$

But $\mathcal{H}_0$ is unknown, so the numerator cannot be observed.

The central knockoff idea is to create **synthetic negative controls** whose behavior can be used to estimate how many selected original variables are null.

The knockoff idea

For every original feature X_j, construct a synthetic feature

$$\widetilde{X}_j, \qquad j = 1, \ldots, p.$$

The knockoff should mimic the relevant dependence structure of the original predictors but should not carry additional information about the response.

We then compare each pair

$$X_j \quad \text{versus} \quad \widetilde{X}_j.$$

If variable j is truly relevant, the original should tend to outperform its knockoff. Under an appropriate null hypothesis, original and knockoff should be statistically interchangeable.

Fixed-X knockoffs

Fixed design linear model

Consider

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2 I_n),$$

where $X \in \mathbb{R}^{n \times p}$ is treated as fixed. The variable-selection target is the support of β.

For the simple direct construction presented below, assume that X has full column rank and, for simplicity, $n \geq 2p$. The original fixed-X framework can also treat $p \leq n < 2p$ by augmenting the design before constructing knockoffs.

In this model define

$$\mathcal{H}_0 = \{j : \beta_j = 0\}, \qquad \mathcal{H}_1 = \{j : \beta_j \neq 0\}.$$

Thus a null variable is a predictor whose coefficient is zero in the fixed-design linear model.

Convention used in this lecture. The design matrix is numeric. We work with centered predictors and, in the numerical examples, scale the columns as well. Centering is mainly a convenient way to handle the intercept separately; it is not an essential knockoff assumption. Categorical predictors can be represented by numerical contrast/dummy columns, although treating a multi-column factor as one scientific variable requires extra care (for example, grouped knockoffs).

Fixed design linear model

Construct $\widetilde{X}$ so that

$$\widetilde{X}^\top \widetilde{X} = X^\top X \quad \text{and} \quad X^\top \widetilde{X} = X^\top X - D,$$

where $D = \mathrm{diag}(s_1, \ldots, s_p)$.

Thus each knockoff has the same Gram structure as the original variables, while its similarity to its own original is deliberately reduced.

Augmented Gram matrix

Let

$$\Sigma = X^\top X.$$

Then

$$[X\ \widetilde{X}]^\top [X\ \widetilde{X}] = \begin{pmatrix} \Sigma & \Sigma - D \\ \Sigma - D & \Sigma \end{pmatrix}.$$

For this matrix to be positive semidefinite we require, in particular,

$$D \succeq 0, \qquad 2\Sigma - D \succeq 0.$$

Here $A \succeq 0$ means that the symmetric matrix A is **positive semidefinite**:

$$v^\top A v \geq 0 \qquad \text{for every vector } v.$$

The choice of D controls how distinguishable originals and knockoffs can be.

Why highly correlated predictors are difficult

If X_j is almost determined by the other predictors, its knockoff must preserve this dependence structure.

Then X_j and $\widetilde{X}_j$ may be very similar.

It becomes difficult for the response to distinguish them.

Strong predictor dependence constrains how different a valid knockoff can be from its original. This can make the original-versus-knockoff competition less informative and reduce power.

Feature statistics

Compare original and knockoff importance

Fit a variable-selection or prediction method using the augmented design

$$[X \ \widetilde{X}] \in \mathbb{R}^{n \times 2p}.$$

For each pair $j = 1, \ldots, p$, define

- Z_j: importance assigned to the original variable X_j;
- $\widetilde{Z}_j$: importance assigned to its knockoff $\widetilde{X}_j$.

We combine the two importances into a **knockoff statistic** W_j. A simple choice is

$$W_j = Z_j - \widetilde{Z}_j.$$

For example, if the augmented model produces coefficients $\widehat{\beta}_1, \ldots, \widehat{\beta}_{2p}$, one may use

$$Z_j = |\widehat{\beta}_j|, \qquad \widetilde{Z}_j = |\widehat{\beta}_{j+p}|.$$

Hence $W_j > 0$ means the original wins, whereas $W_j < 0$ means the knockoff wins.

What is required of the statistic?

For **fixed-X knockoffs**, the classical guarantee uses two properties of the feature statistic:

1. **Sufficiency:** the statistics depend on $(X, \widetilde{X}, Y)$ only through

$$[X\ \widetilde{X}]^\top [X\ \widetilde{X}] \quad \text{and} \quad [X\ \widetilde{X}]^\top Y.$$

2. **Antisymmetry:** swapping X_j and $\widetilde{X}_j$ changes the sign of W_j.

For **model-X knockoffs**, the exchangeability construction is stronger: one uses statistics constructed symmetrically from $(X, \widetilde{X}, Y)$ with the required flip-sign property, without the same fixed-X sufficiency restriction.

The sign-flip principle

For null variables, the original and knockoff should be statistically interchangeable.

Under the corresponding validity conditions, the **sign-flip property** says that, conditional on the magnitudes $|W|$, the signs of the null statistics behave like independent fair coin flips. In particular, for $j \in \mathcal{H}_0$ with $|W_j| > 0$,

$$P(W_j > 0 \mid |W|) = P(W_j < 0 \mid |W|) = \frac{1}{2}.$$

Therefore large negative statistics provide a reference for how many large positive statistics are likely to be null.

This is the key calibration mechanism.

From knockoff statistics to a selected set

Fix a desired FDR level $q \in (0, 1)$. For a candidate threshold $t > 0$, define

$$R(t) = \#\{j : W_j \geq t\}, \qquad L(t) = \#\{j : W_j \leq -t\}.$$

Here $R(t)$ counts sufficiently strong **original wins**, whereas $L(t)$ counts equally strong **knockoff wins**.

For null variables, sign symmetry makes $L(t)$ informative about the unobserved number of false discoveries among the $R(t)$ original wins.

Knockoff+ threshold

Knockoff+ uses the observable quantity

$$\widehat{\mathrm{FDP}}_+(t) = \frac{1 + L(t)}{R(t) \vee 1} = \frac{1 + \#\{j : W_j \leq -t\}}{\#\{j : W_j \geq t\} \vee 1}.$$

It chooses

$$T = \min \left\{ t \in \mathcal{W} : \widehat{\mathrm{FDP}}_+(t) \leq q \right\},$$

where $\mathcal{W} = \{|W_j| : |W_j| > 0\}$. If the set is empty, define $T = \infty$.

Finally select

$$\widehat{S} = \{j : W_j \geq T\}.$$

The extra $+1$ in the numerator is the distinction between the knockoff+ threshold and the original knockoff threshold and provides the finite-sample FDR guarantee under the required assumptions.

FDP, estimated FDP and FDR are different

Now that $R(t)$ and $L(t)$ are defined, we can distinguish three quantities that play different roles:

Quantity	Meaning	Observable?
$\mathrm{FDP} = V/(R \vee 1)$	realized fraction of false discoveries	**No**, because $\mathcal{H}_0$ is unknown
$\widehat{\mathrm{FDP}}_+(t) = (1 + L(t))/(R(t) \vee 1)$	knockoff calibration quantity at threshold t	**Yes**
$\mathrm{FDR} = E(\mathrm{FDP})$	repeated-sampling expectation of FDP	not from one data set

The knockoff method does **not** observe the realized FDP. It uses the observable calibration quantity $\widehat{\mathrm{FDP}}_+(t)$ to choose a threshold, and the theory then gives FDR control under the knockoff assumptions.

Interactive 1

Interactive task 1: calculate a knockoff threshold

Suppose

$$W = (4.8,\ 3.2,\ 2.5,\ 1.7,\ 0.8,\ -0.9,\ -1.8,\ -2.6,\ 0.2,\ -0.1)$$

and the target is

$$q = 0.30.$$

Using the knockoff+ rule, evaluate all distinct positive values

$$t \in \mathcal{W} = \{|W_j| : |W_j| > 0\}.$$

Which variables are selected?

Solution 1: threshold table

For each t, compute

$$\widehat{\mathrm{FDP}}_+(t) = \frac{1 + \#\{W_j \le -t\}}{\#\{W_j \ge t\} \vee 1}.$$

```r
W <- c(4.8, 3.2, 2.5, 1.7, 0.8, -0.9, -1.8, -2.6, 0.2, -0.1)
cand <- sort(unique(abs(W[W != 0])))
tab <- data.frame(
  t = cand,
  neg = sapply(cand, function(t) sum(W <= -t)),
  pos = sapply(cand, function(t) sum(W >= t))
)
tab$fdp_hat <- (1 + tab$neg) / pmax(tab$pos, 1)
```

Solution 1: threshold table

```
tab
```

```
##       t neg pos   fdp_hat
## 1   0.1   4   6 0.8333333
## 2   0.2   3   6 0.6666667
## 3   0.8   3   5 0.8000000
## 4   0.9   3   4 1.0000000
## 5   1.7   2   4 0.7500000
## 6   1.8   2   3 1.0000000
## 7   2.5   1   3 0.6666667
## 8   2.6   1   2 1.0000000
## 9   3.2   0   2 0.5000000
## 10 4.8   0   1 1.0000000
```

For these particular W's and $q = 0.30$, **no finite threshold satisfies knockoff+**.

Solution 1: threshold table

For example, at $t = 2.5$,

$$\widehat{\mathrm{FDP}}_+(2.5) = \frac{1 + \#\{W_j \le -2.5\}}{\#\{W_j \ge 2.5\}} = \frac{1+1}{3} = \frac{2}{3} > 0.30.$$

Even the most favorable candidate in the table remains above 0.30.

Hence

$$T = \infty, \qquad \widehat{S} = \varnothing.$$

This is an important lesson: stringent FDR control may yield no discoveries when the evidence is insufficient.

Model-X knockoffs

Random predictors

Fixed-X knockoffs are tied to a linear-model design.

Model-X knockoffs instead treat

$$X = (X_1, \ldots, X_p)$$

as random. Exact model-X guarantees assume that the distribution P_X is known well enough to generate valid knockoffs. In practice P_X is often estimated, leading to approximate knockoffs.

The response model

$$Y \mid X$$

can be very flexible.

This shifts the modelling burden:

model the predictors well; leave $Y \mid X$ largely unrestricted.

Pairwise exchangeability

A Model-X knockoff $\widetilde{X} = (\widetilde{X}_1, \ldots, \widetilde{X}_p)$ is constructed so that swapping any subset of original/knockoff pairs leaves the joint distribution unchanged.

For $A \subseteq \{1, \ldots, p\}$, let $(X, \widetilde{X})_{\mathrm{swap}(A)}$ mean that X_j and $\widetilde{X}_j$ are exchanged for every $j \in A$. The requirement is

$$(X, \widetilde{X})_{\mathrm{swap}(A)} \stackrel{d}{=} (X, \widetilde{X}) \qquad \text{for every } A.$$

Here $\stackrel{d}{=}$ means equality in distribution.

Also,

$$\widetilde{X} \perp Y \mid X.$$

The knockoff mimics the predictor dependence but is generated without looking at the response.

What is a null variable in Model-X?

Write

$$X_{-j} = (X_1, \ldots, X_{j-1}, X_{j+1}, \ldots, X_p),$$

that is, all predictors except X_j.

In the Model-X framework the null set is

$$\mathcal{H}_0 = \{j : Y \perp X_j \mid X_{-j}\}.$$

Thus variable j is null when, conditional on all other predictors, X_j provides no additional information about Y.

This is a conditional notion of relevance.

A variable may be marginally associated with Y but still be null after conditioning on the other predictors.

Knockoffs therefore target conditional variable importance.

Gaussian model-X construction

If

$$X \sim N(0, \Sigma),$$

one can construct a joint Gaussian pair

$$\begin{pmatrix} X \\ \widetilde{X} \end{pmatrix} \sim N\left(0, \begin{pmatrix} \Sigma & \Sigma - D \\ \Sigma - D & \Sigma \end{pmatrix}\right).$$

Equivalently,

$$\widetilde{X} \mid X$$

is Gaussian with a mean depending on X and a covariance chosen to satisfy exchangeability.

This makes Gaussian knockoffs computationally convenient.

Conditional Gaussian formula

Assuming Σ is invertible,

$$E(\widetilde{X} \mid X) = X - X\Sigma^{-1}D$$

when vectors are represented as row vectors.

The conditional covariance is

$$\mathrm{Var}(\widetilde{X} \mid X) = 2D - D\Sigma^{-1}D.$$

Thus generating knockoffs becomes a multivariate-normal simulation problem.

How can the knockoffs be constructed in practice?

There is not one universal construction. The choice depends first on the statistical setting.

Fixed-X: preserve the sample Gram structure exactly.

- **Equicorrelated knockoffs:** fast and simple; use a common diagonal value in D.
- **SDP knockoffs:** choose D by semidefinite optimization to make originals and knockoffs less correlated, often improving power at greater computational cost.

Model-X: reproduce the predictor distribution.

- **Gaussian knockoffs:** appropriate when a multivariate Gaussian model for X is justified and its parameters are known or estimated.
- **Second-order knockoffs:** match first and second moments; convenient, but only approximate exchangeability outside settings where those moments characterize the distribution.
- More flexible generative methods can be used for complicated predictor distributions, provided the required exchangeability is credible.

The `knockoff` R package implements fixed-X, Gaussian model-X, and second-order constructions. The construction and the feature statistic are separate choices.

Interactive 2

Interactive task 2: what should a knockoff preserve?

Consider three predictors:

- X_1 and X_2 have correlation 0.9;
- X_3 is independent of them;
- Y depends only on X_1.

Discuss:

1. Why is generating $\widetilde{X}_j$ as independent $N(0,1)$ noise generally invalid?
2. What dependence should $\widetilde{X}_1$ reflect?
3. Why might distinguishing X_1 from its knockoff be harder when predictor correlations are very high?
4. Should $\widetilde{X}$ be generated using Y?

Solution 2

Independent noise would fail to reproduce the joint dependence structure of X. An algorithm could then distinguish originals from knockoffs simply because their predictor distributions differ.

Because X_1 is strongly correlated with X_2, $\widetilde{X}_1$ must participate in a joint distribution that preserves the required swap/exchangeability property relative to this correlation structure.

High predictor correlation restricts how different X_1 and $\widetilde{X}_1$ can be while preserving the predictor distribution, reducing power.

Finally,

$$\boxed{\widetilde{X} \text{ must not be constructed using } Y.}$$

In model-X knockoffs the required conditional independence is

$$\widetilde{X} \perp Y \mid X.$$

A simulation workflow

Gaussian correlated predictors

```r
set.seed(14)
n <- 500
p <- 20
rho <- 0.5
Sigma <- outer(seq_len(p), seq_len(p), function(i, j) rho^abs(i - j))
X <- matrix(rnorm(n * p), n, p) %*% chol(Sigma)
beta <- c(3, -2.5, 2, rep(0, p - 3))
y <- as.vector(X %*% beta + rnorm(n, sd = 2))
```

For this simulation the true non-null set is

$$\mathcal{H}_1 = \{1, 2, 3\}.$$

The predictors are correlated, so marginal screening can easily spread apparent signal to neighboring variables.

Generate simple Gaussian knockoffs

For teaching purposes, suppose Σ is known.

Here the predictors are standardized, so $\Sigma_{jj} = 1$. For the equicorrelated construction choose

$$D = sI_p, \qquad s = \min\{1,\, 2\lambda_{\min}(\Sigma)\}.$$

```
s <- min(1, 2 * min(eigen(Sigma, symmetric = TRUE, only.values = TRUE)$values))
D <- diag(s, p)
M <- diag(p) - solve(Sigma, D)
V <- 2 * D - D %*% solve(Sigma, D)
R <- chol(V + 1e-10 * diag(p))
Xk <- X %*% M + matrix(rnorm(n * p), n, p) %*% R
```

This is a transparent Gaussian illustration, not a replacement for a robust knockoff software implementation.

A simple importance statistic

To keep the computation transparent, regress Y on all original and knockoff variables:

$$Y \sim [X, \widetilde{X}].$$

```
fit <- lm(y ~ cbind(X, Xk))
b <- coef(fit)[-1]
Z <- abs(b[seq_len(p)])
Zk <- abs(b[p + seq_len(p)])
W <- Z - Zk
round(W, 2)
```

```
##  cbind(X, Xk)1  cbind(X, Xk)2  cbind(X, Xk)3  cbind(X, Xk)4  cbind(X, Xk)5
##           3.02           2.73           2.13           0.00          -0.11
##  cbind(X, Xk)6  cbind(X, Xk)7  cbind(X, Xk)8  cbind(X, Xk)9 cbind(X, Xk)10
##          -0.12           0.09           0.01           0.19           0.12
## cbind(X, Xk)11 cbind(X, Xk)12 cbind(X, Xk)13 cbind(X, Xk)14 cbind(X, Xk)15
##           0.46           0.21          -0.24           0.01           0.38
## cbind(X, Xk)16 cbind(X, Xk)17 cbind(X, Xk)18 cbind(X, Xk)19 cbind(X, Xk)20
##          -0.10          -0.17          -0.30           0.33           0.18
```

Apply the threshold

```
knockoff_threshold <- function(W, q = 0.2, offset = 1) {
  ts <- sort(unique(abs(W[W != 0])))
  for (t in ts) {
    ratio <- (offset + sum(W <= -t)) / max(1, sum(W >= t))
    if (ratio <= q) return(t)
  }
  Inf
}

T <- knockoff_threshold(W, q = 0.2, offset = 1)
selected <- which(W >= T)
T
```

```
## [1] 0.3332069
```

Apply the threshold

```
selected
```

```
##  cbind(X, Xk)1  cbind(X, Xk)2  cbind(X, Xk)3 cbind(X, Xk)11 cbind(X, Xk)15
##              1              2              3             11             15
## cbind(X, Xk)19
##             19
```

The argument `offset = 1` gives the **knockoff+** rule,

$$\frac{1 + L(t)}{R(t) \vee 1},$$

whereas `offset = 0` corresponds to the original knockoff threshold,

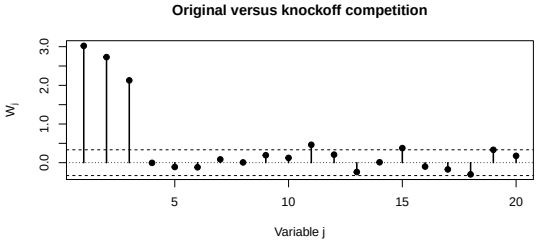
$$\frac{L(t)}{R(t) \vee 1}.$$

The entire inferential procedure is driven by the competition between each feature and its synthetic control.

Visualizing the W_j's and the threshold

```
plot(seq_len(p), W, type = "h", lwd = 2,
     xlab = "Variable j", ylab = expression(W[j]),
     main = "Original versus knockoff competition")
points(seq_len(p), W, pch = 19)
abline(h = 0, lty = 3)
if (is.finite(T)) {
  abline(h = c(-T, T), lty = 2)
}
```

Visualizing the W_j's and the threshold



If $T < \infty$:

- $W_j \geq T$: strong **original wins**, which are selected;
- $W_j \leq -T$: equally strong **knockoff wins**, used for calibration;
- $-T < W_j < T$: not selected.

If $T = \infty$, no variable passes the knockoff+ threshold.

A small real-data example

Why not `mtcars` here?

With only a few predictors, knockoff+ can be quite discrete and conservative. For example, if there are no large negative statistics, at target FDR $q = 0.20$ we still need at least five positive wins before

$$\frac{1 + L(t)}{R(t) \vee 1} \leq 0.20$$

can hold.

So an empty selection for `mtcars` is **not an execution error**; it is actually a useful illustration of the $+1$ correction. For a real-data demonstration in which the procedure is more likely to keep variables, we use a slightly larger classical data set.

Fixed-X knockoffs for the diabetes data

The `lars` package contains the diabetes data from Efron et al. (2004): 442 patients, ten baseline predictors, and a quantitative measure of disease progression one year later. The predictors are already centered and normalized to unit L_2 norm.

For the main worked example we use the **SDP fixed-X construction**. In this realization it produces a nonempty selection, which makes the full knockoff workflow easier to see.

```
library(knockoff)
data(diabetes, package = "lars")
X_diab <- diabetes$x
y_diab <- diabetes$y
# n = 442 and p = 10, so n >= 2p.
# create.fixed() returns a knockoff.variables object containing
# the normalized original design X and the knockoff design Xk.
set.seed(1)
ko_diab <- create.fixed(X_diab, method = "sdp")
X_diab_used <- ko_diab$X
Xk_diab <- ko_diab$Xk
```

Fixed-X knockoffs for the diabetes data

```
# Preserve the predictor names for display.
colnames(X_diab_used) <- colnames(X_diab)
colnames(Xk_diab) <- colnames(X_diab)

# Compute the feature statistics and the knockoff+ threshold.
W_diab <- stat.glmnet_lambdasmax(X_diab_used, Xk_diab, y_diab)
T_diab <- knockoff.threshold(W_diab, fdr = 0.20, offset = 1)
selected_diab <- which(W_diab >= T_diab)
```

Which variables are selected?

```
T_diab
```

```
## [1] 1459.252
```

```
colnames(X_diab_used)[selected_diab]
```

```
## [1] "sex" "bmi" "map" "tc"  "ltg" "glu"
```

In this run the SDP construction selects several variables, so we can see the complete path from knockoff construction to a nonempty selected set.

The purpose is **not** to make a medical claim from this classroom analysis. For fixed-X, `stat.glmnet_lambdasmax` is used because the classical guarantee requires the statistic to satisfy the fixed-X sufficiency property.

What does the augmented design look like?

Because we constructed the knockoff matrix explicitly, we can inspect the actual augmented design used in the analysis. Looking at a **small block** is more informative than printing all 20 columns:

```
# Show three representative original/knockoff pairs.  Use column
# positions so this remains robust to the naming convention of the data set.
show_idx <- c(3, 4, 9)
show_vars <- colnames(X_diab_used)[show_idx]

aug_small <- cbind(
  X_diab_used[, show_idx, drop = FALSE],
  Xk_diab[, show_idx, drop = FALSE]
)
colnames(aug_small) <- c(show_vars, paste0(show_vars, "_ko"))
```

What does the augmented design look like?

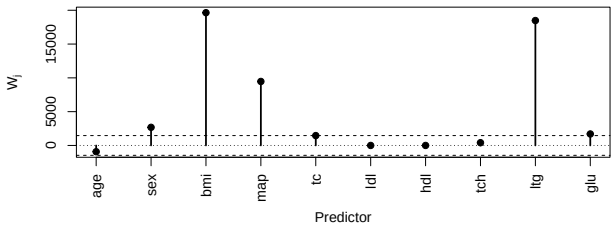
```
round(head(aug_small, 6), 3)
```

```
##          bmi    map    ltg bmi_ko map_ko ltg_ko
## [1,]  0.062  0.022  0.020  0.032  0.110 -0.004
## [2,] -0.051 -0.026 -0.068 -0.042  0.002 -0.043
## [3,]  0.044 -0.006  0.003 -0.055  0.042  0.013
## [4,] -0.012 -0.037  0.023 -0.038 -0.074 -0.005
## [5,] -0.036  0.022 -0.032  0.062 -0.029  0.003
## [6,] -0.041 -0.019 -0.041  0.018 -0.027 -0.053
```

Each row now contains an observed predictor and a synthetic competitor. The individual numerical values are **not** themselves a validity diagnostic; the informative property is the joint Gram/exchangeability structure.

Original versus knockoff statistics in the real data

```
plot(seq_along(W_diab), W_diab, type = "h", lwd = 2,
     xaxt = "n", xlab = "Predictor", ylab = expression(W[j]))
axis(1, at = seq_along(W_diab), labels = colnames(X_diab_used), las = 2)
points(seq_along(W_diab), W_diab, pch = 19)
abline(h = 0, lty = 3)
if (is.finite(T_diab)) abline(h = c(-T_diab, T_diab), lty = 2)
```



This is often more informative than printing the full $[X\ \widetilde{X}]$ matrix: we can immediately see which originals beat their knockoffs strongly enough to cross the knockoff+ threshold.

Compare two fixed-X constructions

Now keep the response, feature statistic and target FDR fixed, but replace the SDP knockoffs by the simpler **equicorrelated** construction.

```
ko_equi <- function(X) create.fixed(X, method = "equi")
set.seed(1)
res_equi <- knockoff.filter(
  X_diab, y_diab, fdr = 0.20,
  knockoffs = ko_equi,
  statistic = stat.glmnet_lambdasmax
)
res <- list(
  SDP = colnames(X_diab_used)[selected_diab],
  equicorrelated = colnames(X_diab)[res_equi$selected]
)
```

Compare two fixed-X constructions

```
res
```

```
## $SDP
## [1] "sex" "bmi" "map" "tc"  "ltg" "glu"
##
## $equicorrelated
## character(0)
```

For these data, the SDP construction keeps variables whereas the equicorrelated construction keeps none. Both are valid fixed-X constructions: they create different synthetic competitors and can therefore have quite different **power** in one realized data set. The FDR guarantee concerns repeated sampling, not equality of the selected sets.

What the package is doing for us

The wrapper `knockoff.filter()` performs the same steps developed manually in this lecture:

$$\boxed{X \longrightarrow \widetilde{X} \longrightarrow W \longrightarrow T \longrightarrow \widehat{S}}$$

The pieces are also available directly:

```
ko <- create.fixed(X_diab, method = "sdp")
X_used <- ko$X
Xk <- ko$Xk
W <- stat.glmnet_lambdasmax(X_used, Xk, y_diab)
T <- knockoff.threshold(W, fdr = 0.20, offset = 1)
selected <- which(W >= T)
```

This is useful pedagogically: the package automates the mechanics, but the objects correspond directly to the construction, statistic and threshold already introduced.

Why the guarantee works

Negative statistics estimate false positives

For null features, signs are symmetric.

At threshold t,

- $\#\{W_j \geq t\}$ contains discoveries;
- $\#\{W_j \leq -t\}$ acts as a proxy for false discoveries among them.

Hence

$$\frac{1 + \#\{W_j \leq -t\}}{\#\{W_j \geq t\} \vee 1}$$

is an observable calibration quantity for the unobserved false-discovery proportion. The knockoff+ theorem uses the null sign-flip property to turn this calibration into FDR control; it should not be interpreted as the actual FDP of the realized selected set.

No knowledge of the unknown active set is required.

Why antisymmetry matters

Suppose we swap X_j and $\widetilde{X}_j$.

A valid statistic should satisfy

$$W_j \longrightarrow -W_j$$

while the other relevant statistics transform consistently.

This makes the sign of W_j encode which member of the pair won.

A statistic that treats the original and knockoff asymmetrically by construction destroys the calibration argument.

Interactive 3

Interactive task 3: design a valid statistic

Suppose a fitted model gives coefficient magnitudes

$$Z_j = |\widehat{\beta}_j|, \qquad \widetilde{Z}_j = |\widehat{\beta}_{j+p}|.$$

Consider:

A. $W_j = Z_j$

B. $W_j = Z_j - \widetilde{Z}_j$

C. $W_j = \max(Z_j, \widetilde{Z}_j)$

D. $W_j = (Z_j - \widetilde{Z}_j)\max(Z_j, \widetilde{Z}_j)$

Which statistics change sign when the original and knockoff are swapped? Which could therefore serve as knockoff statistics?

Solution 3

After swapping,

$$Z_j \leftrightarrow \widetilde{Z}_j.$$

A. Z_j becomes $\widetilde{Z}_j$, not $-Z_j$: invalid.

B.

$$Z_j - \widetilde{Z}_j \longrightarrow \widetilde{Z}_j - Z_j = -W_j :$$

valid.

C. The maximum is unchanged: invalid.

D. The difference changes sign while the maximum is unchanged, so

$$W_j \longrightarrow -W_j :$$

valid.

Thus B and D have the required antisymmetry.

FDR versus power

FDR control does not mean perfect selection

Even with valid knockoffs:

- some active variables may be missed;
- some null variables may be selected;
- correlated predictors can reduce power;
- small samples can yield no discoveries.

The guarantee concerns the expected false-discovery proportion, not exact recovery of the active set.

Statistical validity and statistical power are different questions.

The quality of model-X knockoffs matters

Exact model-X guarantees assume a valid knockoff distribution.

In practice P_X is usually estimated.

If the estimated predictor distribution is poor:

- exchangeability is only approximate;
- FDR control may deteriorate;
- synthetic variables may be too easy to distinguish from originals.

This creates a new computational-statistical problem: learning P_X well enough to construct useful negative controls.

Interactive 4

Interactive task 4: empirical FDP and power

In one simulation the true non-null set is

$$\mathcal{H}_1 = \{1, 2, 3, 4, 5\}$$

and a knockoff procedure selects

$$\widehat{S} = \{1, 2, 4, 7\}.$$

1. Compute the FDP.
2. Compute the power, defined here as the proportion of active variables selected.
3. If the target FDR was $q = 0.20$, does this single realization prove that FDR control failed?
4. What experiment would be appropriate for assessing FDR empirically?

Solution 4

There is one false discovery:

$$V = 1$$

and four total discoveries:

$$R = 4.$$

Therefore

$$\text{FDP} = \frac{1}{4} = 0.25.$$

Three of the five active variables were selected, so

$$\text{power} = \frac{3}{5} = 0.60.$$

A single FDP of $0.25 > 0.20$ does **not** contradict FDR control because

$$\text{FDR} = E(\text{FDP})$$

is an expectation over repeated data sets/procedure randomness.

To assess it empirically, repeat the complete simulation many times and average the realized FDPs.

Empirical FDR and power in repeated simulations

The following code repeats the **same Gaussian model-X workflow** and records FDP and power.

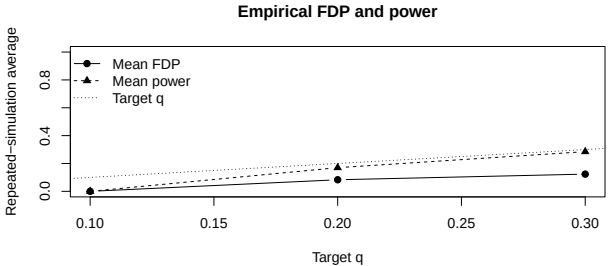
```
one_run <- function(q, n, Sigma, beta, s) {
  p <- length(beta)
  Xr <- matrix(rnorm(n * p), n, p) %*% chol(Sigma)
  yr <- as.vector(Xr %*% beta + rnorm(n, sd = 2))
  D <- diag(s, p)
  M <- diag(p) - solve(Sigma, D)
  V <- 2 * D - D %*% solve(Sigma, D)
  Rchol <- chol(V + 1e-10 * diag(p))
  Xkr <- Xr %*% M + matrix(rnorm(n * p), n, p) %*% Rchol
  fitr <- lm(yr ~ cbind(Xr, Xkr))
  br <- coef(fitr)[-1]
  Wr <- abs(br[seq_len(p)]) - abs(br[p + seq_len(p)])
  Tr <- knockoff_threshold(Wr, q = q, offset = 1)
  Sr <- which(Wr >= Tr)
  active <- which(beta != 0)
  Vfalse <- sum(!(Sr %in% active))
  fdp <- Vfalse / max(length(Sr), 1)
  power <- sum(Sr %in% active) / length(active)
  c(FDP = fdp, Power = power)
}
```

Empirical FDR and power in repeated simulations

```
set.seed(1401)
qs <- c(0.10, 0.20, 0.30)
B <- 200
res <- lapply(qs, function(q) {
  replicate(B, one_run(q, n = 500, Sigma = Sigma, beta = beta, s = s))
})
mean_fdp <- sapply(res, function(z) mean(z["FDP", ]))
mean_power <- sapply(res, function(z) mean(z["Power", ]))

plot(qs, mean_fdp, type = "b", pch = 19, ylim = c(0, 1),
     xlab = "Target q", ylab = "Repeated-simulation average",
     main = "Empirical FDP and power")
lines(qs, mean_power, type = "b", pch = 17, lty = 2)
abline(a = 0, b = 1, lty = 3)
legend("topleft", c("Mean FDP", "Mean power", "Target q"),
       pch = c(19, 17, NA), lty = c(1, 2, 3), bty = "n")
```

Empirical FDR and power in repeated simulations



The mean FDP estimates the FDR across repetitions. The point is not that every realized FDP lies below q, but that the **average** should be controlled when the knockoff assumptions hold. Power is a separate performance criterion.

Connections

Knockoffs and predictor augmentation

Both methods deliberately add artificial variables.

Predictor augmentation:

- artificial variables are known-noise directions;
- their eigenvector coordinates help reveal structural rank/order.

Knockoffs:

- artificial variables mimic the dependence structure of the real predictors;
- pairwise competition calibrates variable selection.

The synthetic variables are designed for different inferential purposes.

Knockoffs and permutation ideas

There is also a conceptual connection to permutation tests.

Both construct a reference based on symmetry:

- permutation tests: exchangeability under the null generates a reference distribution;
- knockoffs: original/knockoff exchangeability makes null signs symmetric.

In both cases, validity comes from designing a transformation whose behavior is known under the null.

Knockoffs and simulation

Knockoff generation is a simulation problem.

The full pipeline is

$$\text{estimate/know } P_X \to \text{simulate } \widetilde{X} \mid X \to \text{fit augmented model} \to W_j \to \text{FDR threshold.}$$

Thus random-number generation, linear algebra, modelling, optimization and multiple-testing ideas all meet in one procedure.

Practical workflow

A checklist

1. Decide whether fixed-X or model-X formulation is appropriate.
2. Construct valid knockoffs.
3. Check basic diagnostics of original/knockoff similarity and dependence.
4. Fit a model to $[X, \widetilde{X}]$.
5. Construct valid W_j's: for fixed-X, check sufficiency and antisymmetry; for model-X, check the required flip-sign symmetry.
6. Choose the target FDR q.
7. Apply the knockoff or knockoff+ threshold.
8. Report selected variables and the precise inferential target.
9. Discuss power and the assumptions behind knockoff construction.

What not to do

Do not:

- generate independent random noise and call it a knockoff;
- use the response when constructing model-X knockoffs;
- choose a statistic that inherently favors originals;
- interpret FDR q as saying every selected set has FDP $\leq q$;
- ignore dependence among predictors;
- treat approximate knockoffs as exact without discussing the predictor model.

References and summary

Main references

- Barber, R. F. and Candès, E. J. (2015). *Controlling the false discovery rate via knockoffs*. **The Annals of Statistics**, 43(5), 2055–2085. https://doi.org/10.1214/15-AOS1337.
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- Benjamini, Y. and Hochberg, Y. (1995). *Controlling the false discovery rate: a practical and powerful approach to multiple testing*. **Journal of the Royal Statistical Society: Series B**, 57(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.
- Efron, B., Hastie, T., Johnstone, I. and Tibshirani, R. (2004). *Least angle regression*. **The Annals of Statistics**, 32(2), 407–499. https://doi.org/10.1214/009053604000000067.

Take-home messages

1. Knockoffs create **synthetic negative controls** for variable selection.
2. Valid knockoffs mimic the dependence structure of the predictors.
3. Under the appropriate fixed-X or model-X conditions, null original/knockoff pairs yield the sign-flip principle.
4. Negative W_j's calibrate the number of false positive W_j's.
5. Knockoff+ converts this calibration into finite-sample FDR control under its assumptions.
6. Model-X knockoffs target conditional relevance $Y \not\perp X_j \mid X_{-j}$.
7. FDR control is not exact support recovery; power and predictor dependence still matter.
8. Together with predictor augmentation, knockoffs show how **carefully designed artificial data can enable inference**.